BRANCHES OF A NOETHERIAN LOCAL RING VERSUS ITS COMPLETION

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Allowed standard prerequisites (editorial).

- The map from the henselization to the completion of a Noetherian local ring is flat and injective. invoked at proof lines 31949-31950. Sets up the minimal prime comparison.
- Flat maps satisfy going down and injective ring maps realize every minimal prime of the target as lying over one from the source. invoked at proof lines 31950-31952. Proves part (1).
- The completion is henselian and the number of branches is defined via minimal primes of henselizations. invoked at proof lines 31953-31956. Proves part (2).
- The strict henselization of the completion is the base change of the strict henselization over the henselization. invoked at proof lines 31957-31958. Enables repeating the argument for part (3).

Source reference. Original lemma 06LJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10979–10979. Cited at proof lines 31949-31949 and 31957-31958 for the flat injective map and the strict henselization base change; the minimal prime and branch arguments are inside the counted span.

Source reference. Original lemma 00HS: sources/stacks-chow/tags.txt, source lines 654–654. Cited at proof lines 31950-31951 as the first ingredient for part (1); the conclusion is inside the counted span.

Source reference. Original lemma 00FK: sources/stacks-chow/tags.txt, source lines 574–574. Cited at proof lines 31951-31952 as the second ingredient for part (1); the conclusion is inside the counted span.

Source reference. Original definition 0C26: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31553–31553. Cited at proof lines 31954-31954 for the definitions behind parts (2) and (3); the inequalities are proved inside the counted span.

Source reference. Original lemma 04GM: sources/stacks-chow/tags.txt, source lines 5838–5838. Cited at proof lines 31955-31956 to identify the branches of the completion; the inequality is inside the counted span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring.*

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(1) *The map $A^h \to A^\wedge$ defines a surjective map from minimal primes of $A^\wedge$ to minimal primes of A^h.*
(2) *The number of branches of A is at most the number of branches of $A^\wedge$.*
(3) *The number of geometric branches of A is at most the number of geometric branches of $A^\wedge$.*

Proof. By Lemma 06LJ the map $A^h \to A^\wedge$ is flat and injective. Combining going down (Algebra, Lemma 00HS) and Algebra, Lemma 00FK we see that part (1) holds. Part (2) follows from this, Definition 0C26, and the fact that $A^\wedge$ is henselian (Algebra, Lemma 04GM). By Lemma 06LJ we have $(A^\wedge)^{sh} = A^{sh} \otimes_{A^h} A^\wedge$. Thus we can repeat the arguments above using the flat injective map $A^{sh} \to (A^\wedge)^{sh}$ to prove (3). □